# A Novel Adaptive Hybrid Fusion Network for Multiresolution Remote Sensing Images Classification

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***Abstract*— With the rapid development of earth observation technology, panchromatic (PAN) and multispectral (MS) images have also become easier to obtain. The multiresolution classification of PAN and MS images as a basic MS image analysis task has become a research hotspot. The main challenge in this field is how to process data and extract features to improve classification accuracy effectively. In this article, we design a novel adaptive hybrid fusion network (AHF-Net) for multiresolution remote sensing image classification. It includes two parts: data fusion and feature fusion. In the data fusion part, we propose an adaptive weighted intensity-hue-saturation (AWIHS) strategy, which can reduce the difference between MS and PAN images by adaptively adding each other's unique information from the perspective of information sharing. In the feature fusion part, starting from the second-order correlation of features, we propose a correlation-based attention feature fusion (CAFF) module. It can improve the discrimination of fusion features by adaptively determining the fusion coefficient according to the importance of the input feature channel. Based on AWIHS and CAFF, inspired by the idea of feature pyramid, we combine the multilevel feature fusion and the dual-branch residual network as the backbone network of AHF-Net. By combining AWIHS and CAFF modules with the backbone network, our AHF-Net can effectively improve the classification accuracy of multiresolution remote sensing images. The effectiveness of the proposed algorithm has been verified on multiple data sets. Our code and model are available at https://github.com/1826133674/AHF-Net.**

***Index Terms*— Data difference reduction, deep learning (DL), feature fusion, multiresolution image classification, remote sensing.**

## I. Introduction

NOWADAYS, with the continuous development of earth observation technology, the performance and quantity of remote sensing images have been greatly improved. However, at the current level of technology, it is still impossible to simultaneously obtain remote sensing images with high spectral information and high spatial resolution. A few years ago, space-borne passive earth observation systems such as QuickBird, Worldview, Pleiades, Spot5 HRG, DEMIMOS-2, IKONOS, Landsat 7 ETM+, and Landsat 8 OLI, can jointly acquire two images of the same scene, i.e., a panchromatic (PAN) image with high spatial resolution but low spectral resolution and a multispectral (MS) image with high spectral resolution but low spatial resolution. The intrinsic relationship between these two images lays the foundation for the multiresolution classification of PAN and MS images [1]. Compared with pure MS image classification, the goal of PAN and MS image multiresolution classification is to use their internal consistency to extract and fuse features to classify MS images accurately based on reducing their external differences. PAN and MS multiresolution classification algorithms can be roughly divided into two categories: One is performing data fusion first and then classifying the pansharpening MS image [2]–[6], and the other is extracting the features of MS and PAN separately and then fusing the features for classification [7]–[10].

The former algorithm (data fusion before classification) extracts features from the fused PAN and MS image obtained by pansharpening algorithms and then fuses them for classification. In recent years, many excellent pansharpening algorithms have been proposed. They can be roughly divided into four categories [11]: component substitution (CS) (including: intensity-hue-saturation (IHS) transform [12]–[16], Gram-Schmidt (GS) [17], principal component analysis (PCA) [18]); multiresolution analysis (MRA) [19]; variational optimization (VO) [20]; and deep learning (DL) [21]. Shackelford and Davis [3] used a color normalization method to pansharpening the MS image and then used a fuzzy logic methodology to classify the pansharpening MS image. Su *et al.* [4] used textural and local spatial statistical information to classify the pansharpening MS image. Shingare *et al.* [5] evaluated multiple pansharpening methods and used the decision tree algorithms for pansharpening MS image classification. The above algorithms all have promising performance and provide some useful inspiration for the development of multiresolution classification. However, their classification performance relies heavily on the pansharpening algorithm.

The later algorithm (feature fusion before classification) extracts features from PAN and MS image, respectively, and then fuses the features for classification. Moser and

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Serpico [7] proposed a novel contextual method based on Markov random fields for the multiresolution classification of PAN and MS images. Moser *et al.* [8] used a similar method for the multiresolution classification of PAN and MS images. Under a unified Bayesian framework, Mao *et al.* [9] generated semantic segments from PAN images and used MS images to assign cluster labels to the segments to do the classification. Zhang *et al.* [10] constructed a middle-level representation of the input data and then used the support vector machine (SVM) for image classification. Although these algorithms also extract the features of the PAN and MS images, respectively, they did not consider the significant external differences between PAN and MS data well.

Recently, DL has been widely used in the field of remote sensing [22]–[28]. These methods have gradually become mainstream due to their excellent performance. Inspired by it, researchers achieved the following research results: Liu *et al.* [22] proposed a two-branch classification network, which uses SAE and DCNN to extract the features of MS images and PAN images respectively and then uses a fully connected network to fuse the cascaded features for final classification. Zhao *et al.* [23] proposed a two-stage classification algorithm. First, superpixel segmentation was performed on the MS image, and multiple CNNs were used to extract features of different parts of the superpixel to complete the classification. Then, the information of the PAN image was used to fine-tune the classification result. The proposal of the dual-branch network provides new inspiration for the multiresolution classification of PAN and MS images. Bergado *et al.* [25] proposed a multiresolution convolution network called Fusenet for VHR satellite image classification, which combines band fusion and classification into an end-to-end network by convolution and upsampling. In [26], based on the Fusenet [25], they further propose a network called Reusenet for VHR image classification, which combines the fusion of different resolutions images and regularization into an end-to-end network through the recurrent architecture of ReuseNet. Zhu *et al.* [27] used the spatial attention module and the channel attention module to extract the features of the PAN and MS images, respectively, then fuses the extracted features for classification. The above networks and algorithms have enriched the technology of multiresolution classification and improved the accuracy of multiresolution classification, which inspires us to further mine and use PAN and MS data.

Although the application of DL methods in this field has achieved impressive performance, some easily ignored problems are worth our attention:

1) For data with significant differences, the IHS pansharpening algorithm usually causes spectral distortion, which cannot fuse the characteristic information of the data well. Besides, in order to obtain better fusion results, the IHS pansharpening algorithm usually has some hyperparameters, which reduces the universality and generalization of the algorithm.
2) For the features to be fused, the weighting coefficients of features currently used in feature fusion strategies (such as element-wise adding or feature concatenating) are the same. They are relatively simple and crude, which does not take into account that the actual remote sensing terrains have different preferences for the two input channels. This usually results in the output features cannot inherit the discriminative properties of the input features well.
3) For the framework based on data fusion, feature extraction is performed manually, which cannot extract high-level semantic information in the data well. For feature-based fusion frameworks, there is no difference reduction process for data with two different attributes. They independently extract the characteristics of PAN and MS. Even if they are the terrain of the same scene; the extracted features may not be in the same feature space, thereby increasing the difficulty of feature fusion and the burden of network training.

Based on these, we propose a novel adaptive hybrid fusion network (AHF-Net) for multiresolution remote sensing image classification. Our main contributions are as follows.

1) We propose an adaptive weighted IHS (AWIHS) data fusion strategy to reduce data differences. By using the linear factor to adjust the ratio among the spectral bands, the spectral distortion of the IHS algorithm can be alleviated. And from the perspective of complementary information, the MS image's unique information is added to the PAN image, thereby reducing the difference between the data. Besides, AWIHS has no hyperparameters, and the algorithm has better universality and generalization.
2) We propose a correlation-based attention feature fusion (CAFF) module in the network. This module takes feature correlation as a measure of feature redundancy and generates a corresponding mask to enhance or suppress the channel of fused features. Through this, The CAFF module can adjust the fusion coefficient adaptively by measuring the importance of the channel of the fusion feature, thereby improving the discrimination of the fusion feature.
3) We propose a dual-path hybrid classification framework that combines data fusion and feature fusion. This framework can not only extract features through branch networks but also reduce the difference in feature space through data fusion and the interaction of two branch features. This interaction can make feature fusion more efficient and reliable, thereby improving classification performance.

The rest of the article is organized as follows. Section II introduces the work related to this article's method. Section III elaborates the proposed method in detail. Section IV introduces the details of data sets used in this article and the experimental setup, and then shows the experimental results and the corresponding analysis. Finally, Section V concludes this article.

## II. Related Work

In this section, two works related to our algorithm will be introduced in detail, including the IHS pansharpening algorithms and the second-order global pooling.

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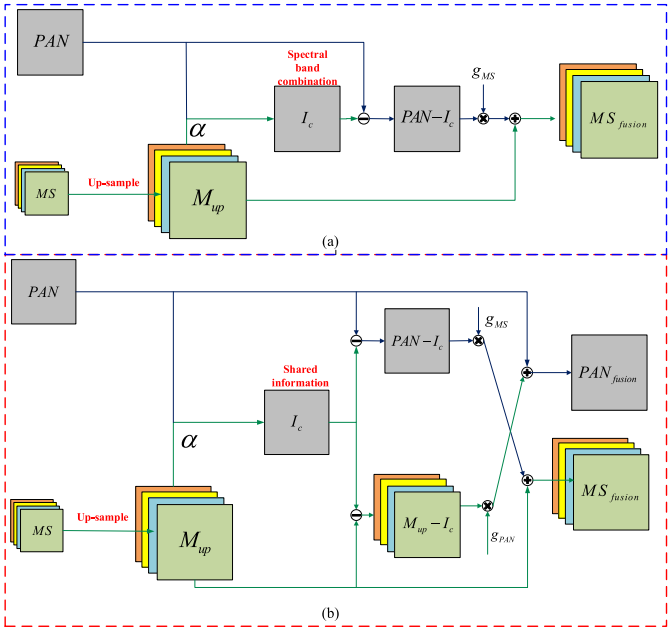



Fig. 1. (a) Flowchart of AIHS pansharpening algorithm. (b) Flowchart of AWIHS algorithm. $\alpha$: linear factor. $I_c$: linear combination of $M_{\text{up}}$ image bands. $g_{\text{MS}}$: fusion coefficient of MS image. $g_{\text{PAN}}$: fusion coefficient of PAN image.

### A. IHS Pansharpening Methods

The pansharpening of MS image refers to adding the spatial details of a PAN image into the MS image. The purpose of sharpening is to obtain MS images with high spectral resolution and high spatial resolution. As mentioned in Introduction, the IHS method is one of the CS pansharpening methods [18], [29]. The original IHS method is very simple and is only suitable for three bands of MS images. The basic flow of the IHS-based pansharpening methods is as shown in Fig. 1(a): First, the MS image is up-sampled to the PAN image size, obtaining the $\boldsymbol{M}_{\text{up}}$. Second, the PAN and $\boldsymbol{M}_{\text{up}}$ image are mapped to IHS color space. And then replace the intensity component $\boldsymbol{I_c}$ of the $\boldsymbol{M}_{\text{up}}$ image with the intensity component of the PAN image. Finally, map the $\boldsymbol{M}_{\text{up}}$ image back to RGB space to obtain the fused image $\boldsymbol{MS}_{\text{fusion}}$. However, since most MS images consist of four channels, RGB and near-infrared, the researchers extended the IHS algorithm to apply to multichannel images [14], [30]. Its expression is as follows:

$$\boldsymbol{MS}^{\boldsymbol{i}}_{\text{fusion}} = \boldsymbol{M}^{\boldsymbol{i}}_{\text{up}} + \boldsymbol{g_{MS}}(\boldsymbol{PAN} - \boldsymbol{I_c}), \quad i = 1, \ldots, n \tag{1}$$

where $\text{MS}^{\boldsymbol{i}}_{\text{fusion}}$ and $\boldsymbol{M}^{\boldsymbol{i}}_{\text{up}}$ are the $i$th band of the $\text{MS}_{\text{fusion}}$ and $\boldsymbol{M}_{\text{up}}$ image, respectively and $\boldsymbol{g_{MS}}$ is the fusion coefficient, $n$ is the number of bands in the MS image. The intensity component $\boldsymbol{I_c}$ is a linear combination of $\boldsymbol{M}_{\text{up}}$, and its calculation method is as follows:

$$I_c = \sum_{i=1}^{n} \alpha_i \boldsymbol{M}^{\boldsymbol{i}}_{\text{up}} \tag{2}$$

where $\alpha_i$ is the linear factor. In the early algorithm, the linear factor and the fusion coefficient were simply set to the constant $(1/n)$. This set will result in severe spectral distortion in the fused image, when the PAN and MS images are significantly different.

In order to reduce the spectral distortion of the fused image, Rahmani *et al.* [13] proposed an adaptive IHS (AIHS) pan-sharpening method for calculating $I_c$. This method obtained the linear factor $\alpha_i$ by optimizing an equation. In order to better add spatial details to the MS image, they also used an edge detection operator to extract the edge information of the PAN image. Based on this, Leung *et al.* [15] proposed an improved AIHS (IAIHS) method. They introduced the spectral information ratio and the edge detection information of the MS image into the fusion coefficient to reduce the spectral distortion further. Based on the AIHS and IAIHS algorithms, Wang [16] proposed an improved algorithm. They introduced a manually adjusted parameter t to adjust the spectral information ratio and used the $\alpha$ as the combination coefficient of the PAN and MS edge detection operators.

The IHS pansharpening algorithms are fast and convenient to implement, and the spatial resolution of the fused image is high too. However, their shortcomings are also obvious. On the one hand, the problem of spectral distortion still exists. On the other hand, the above algorithms need to manually adjust some parameters to achieve a good fusion effect for the images of different sensors.

### B. Global Second-Order Pooling

The concept of global second-order pooling (GSOP) [31] is relative to global first-order pooling (GFOP) [32]. GFOP refers to the average pooling of the entire feature map. In recent years, GFOP has two uses; one is replacing the fully connected layer of the neural network [32]–[34], and the other is squeezing features in the attention mechanism [35]–[37].

GSOP uses covariance for the global pooling of feature maps. The specific process is as follows. Let $\boldsymbol{F} \in \mathbb{R}^{H\times W\times C}$ be the feature map to be pooled, reshape $\boldsymbol{F}$ into $\boldsymbol{F'} \in \mathbb{R}^{HW\times C}$, then calculate the covariance between the channels to get the covariance matrix $\boldsymbol{C} \in \mathbb{R}^{C\times C}$. Matrix C is the result of the feature map $\boldsymbol{F}$'s GSOP.

The study of second-order pooling originates from researchers' research on strengthening the nonlinear capabilities of deep neural networks [31]. Li *et al.* [31] studied the effect of second-order statistics on the performance of large-scale visual recognition tasks in DL. They propose a matrix power normalization method to solve the problems of feature robustness and covariance structure preservation. Wang *et al.* [38] also proposed a Global gated mixture of second-order pooling for improving deep neural networks. Wang *et al.* [39], Li *et al.* [40], and Xiao *et al.* [41] explored the use of second-order pooling at the end of neural networks, that is, using second-order pooling instead of the fully connected layer, and achieved good results. Inspired by the studies [39]–[41], GSOP-Net [42] is inserted into the neural network as a module to explore how to introduce higher-order representations in the early layers of the network to improve the nonlinearity of the network. The exploration of GSOP-Net [42] is to use GSOP to obtain the feature representation in the attention mechanism, which is consistent with the application of GFOP in the attention mechanism.

Compared with the GFOP, the research of GSOP improves the nonlinear capability of neural networks. The extracted

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feature representation information is more abundant and can be easy to operate.

## III. Proposed Method

In this section, the proposed method is introduced in detail, including an AWIHS data fusion strategy, a CAFF block, and the AHF-Net.

### A. AWIHS Data Fusion Strategy

During the acquisition process of PAN and MS images, the sensor and atmospheric medium are different. After geometric and radiation correction, the external differences between the two images are enormous. The relationship between MS and PAN images is: the essential features are the same, but the apparent features are different. This difference usually causes the extracted features to fail to reflect the internal consistency of the two images. If the original MS image and PAN image are directly used as the input of the network, the weight of the network cannot be shared. The features extracted by the network are very likely in different feature spaces, leading to unsatisfactory results of feature fusion. Therefore, the difference between PAN and MS images should be reduced to reduce the burden of subsequent network integration and classification.

For algorithms that use DL to classify MS images and PAN images, they did not consider reducing the data difference between the MS and PAN data before inputting the images into the network. Therefore, we propose a data fusion strategy to reduce the difference between them. We call it an AWIHS image fusion algorithm. The basic flow of AWIHS is shown in Fig. 1(b). The AWIHS takes MS and PAN images as input. It can be divided into two parts: the acquisition of the $\mathrm{MS}_{\mathrm{fusion}}$ image and $\mathrm{PAN}_{\mathrm{fusion}}$ image.

*1) Acquisition of the $MS_{fusion}$ Image:* For an MS image, in order to better interact with the PAN image, we upsample it to the size of the PAN image and obtain the upsampled image: $\boldsymbol{M}_{\mathrm{up}}$. The upsampling method is the bicubic interpolation algorithm. This upsampling method uses more neighborhood information than nearest neighbor interpolation and quadratic interpolation.

We calculate the linear factor $\alpha_i$ by optimizing the following equation:

$$\min_{\alpha_1,\ldots,\alpha_n} = \|\boldsymbol{PAN} - \boldsymbol{I_c}\|^2 = \|\boldsymbol{PAN} - \sum_{i=1}^{n} \alpha_i \boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}}\|^2 \quad \text{s.t. } \alpha_1 \geq 0, \ldots, \alpha_n \geq 0. \tag{3}$$

And then, we calculate $\boldsymbol{I_c}$ according to (2). We consider the $\boldsymbol{I_c}$ from the perspective of sharing information. Since $(\boldsymbol{PAN} - \boldsymbol{I_c}) > 0$, we infer that the information of $\boldsymbol{I_c}$ should be included in the PAN image. Since $0 < \alpha_i < 1$, we infer that the information of $\boldsymbol{I_c}$ should also be included in the MS image. Based on this, the $\boldsymbol{I_c}$ can be regarded as shared information of the MS and PAN images. The difference between the PAN image and $\boldsymbol{I_c}$ can be regarded as the unique information of the PAN image.

In order to obtain the edge information of PAN and MS images, we use edge detection operators $\boldsymbol{\mathcal{W}_{PAN}}$ and $\boldsymbol{\mathcal{W}_{M_{\mathrm{up}}^i}}$ to extract the respective edges. The calculation of the edge detection operators is as shown in the following equation:

$$\boldsymbol{\mathcal{W}_{PAN}} = \exp\left(-\frac{\lambda}{|\nabla \boldsymbol{PAN}|^4 + \varepsilon}\right) \tag{4}$$

$$\boldsymbol{\mathcal{W}_{M_{\mathrm{up}}^i}} = \exp\left(-\frac{\lambda}{|\nabla \boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}}|^4 + \varepsilon}\right) \tag{5}$$

where $\nabla \boldsymbol{PAN}$ and $\nabla \boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}}$ are the gradient of the PAN image and the $i$th channel of the MS image, respectively. $\lambda = 10^{-9}$ is a parameter indicating how large the gradient should be and controls the smoothness of the image, and $\varepsilon = 10^{-10}$ is a small value that enforces a nonzero denominator. Since the edges appearing in the PAN image may not appear in each MS band, it is not suitable to inject the same amount of detail into different bands. We believe that the $\alpha_i$ can reflect the degree of difference between each MS spectral band and the PAN image. Therefore, in order to make the distribution of spatial details in each band of $\mathrm{MS}_{\mathrm{fusion}}$ more reasonable, instead of using $\boldsymbol{\mathcal{W}_{PAN}}$ singly, we use $(1 - \alpha_i)$ and $\alpha_i$ as the weight of the $\boldsymbol{\mathcal{W}_{PAN}}$ and $\boldsymbol{\mathcal{W}_{M_{\mathrm{up}}^i}}$, respectively.

In order to reduce the spectral distortion, the information injected into each band needs to maintain a similar spectral ratio as the MS original band. We consider that $\alpha_i$ can reflect the degree of difference between different spectral bands of MS too. Therefore, $\alpha_i$ can also be used as the adaptive weights to adjust the information injection ratio of each spectral band.

Based on the above analysis, we design the following fusion coefficient $\boldsymbol{g_{MS}}$:

$$\boldsymbol{g_{MS}} = \left(\frac{\alpha_i}{\sum_{i=1}^{n} \alpha_i}\right)\left((1 - \alpha_i)\boldsymbol{\mathcal{W}_{PAN}} + \alpha_i \boldsymbol{\mathcal{W}_{M_{\mathrm{up}}^i}}\right) \tag{6}$$

where $\alpha_i / (\sum_{i=1}^{n} \alpha_i)$ is the modulation parameters. We get the final fusion image $\mathrm{MS}_{\mathrm{fusion}}$ through (1).

The $\alpha_i / (\sum_{i=1}^{n} \alpha_i)$ can adaptively reduce the spectral distortion without the need to adjust similar parameters manually, thus the efficiency and performance of the algorithm have been improved.

*2) Acquisition of $PAN_{fusion}$ Image:* In order to reduce their differences, we start from the perspective of information sharing. For $\mathrm{PAN}_{\boldsymbol{fuison}}$ image, the $\boldsymbol{I_c}$ is also regarded as the shared information of the MS and PAN images. The difference between the MS image and $\boldsymbol{I_c}$ can be regarded as the unique information of the MS image. Since the $\alpha_i$ can reflect the degree of difference between each MS spectral band and the PAN image, we also use $\alpha_i / (\sum_{i=1}^{n} \alpha_i)$ as the weight of the information of different bands of MS injected into the PAN image. We get the $\mathrm{PAN}_{\mathrm{fusion}}$ image by the following equation:

$$\mathrm{PAN}_{\mathrm{fusion}} = \boldsymbol{PAN} + \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \left(\frac{\boldsymbol{\alpha_i}}{\sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{\alpha_i}}\right)\left(\boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}} - \boldsymbol{\alpha_i} \boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}}\right). \tag{7}$$

When $\alpha_i$ is equal to 1/4, the above formula degenerates to the following equation:

$$\mathrm{PAN}_{\mathrm{fusion}} = \boldsymbol{PAN} + \frac{1}{4}\sum_{\boldsymbol{i=1}}^{\boldsymbol{n}}\left(\boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}} - \boldsymbol{\alpha_i} \boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}}\right) = \boldsymbol{PAN} + \frac{1}{4}\left(\sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{M}_{\mathrm{up}}^{\boldsymbol{i}} - \boldsymbol{I_c}\right). \tag{8}$$

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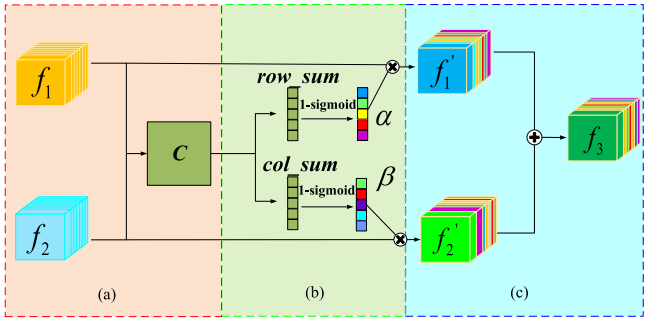



Fig. 2. Flowchart of CAFF module. It can be divided into three parts. (a). Calculation of the correlation matrix $\mathcal{C}$. (b) Calculation of feature fusion Mask $\alpha$ and $\beta$. (c) Acquisition of the fusion feature $f_3$. $f_1$, $f_2$: features to be fused.

The AWIHS mainly has two advantages: On the one hand, the introduction of $\alpha_i/(\sum_{i=1}^{n}\alpha_i)$ can adaptively reduce the spectral distortion caused by fixed parameters and thus increase the effectiveness of the algorithm. On the other hand, from the perspective of information sharing, the AWIHS algorithm separates the shared information of PAN and MS images and adds their unique information to each other. Through this, our AWIHS can adaptively reduce the difference between PAN and MS.

### *B. CAFF Module*

Recently, with a large-scale application of CNN, the structure of network models has become increasingly complex, thus the output features have become increasingly diverse. The scale of remote sensing images terrain often has large differences. The general approach is to use feature element-wise adding or feature concatenating to fuse features of different scales. However, such a feature fusion method is too simple and does not take into account the correlation between features. As a result, the fused features are not effective enough, which reduces the accuracy of classification. In this article, after the AWIHS algorithm, the difference between PAN and MS data is reduced and the internal connection is strengthened. Considering the inherent consistency between features, we propose a CAFF block. The details of the block are as follows.

The entire process of CAFF is as shown in Fig. 2. Suppose the features to be fused are $f_1$, $f_2 \in \mathbb{R}^{H\times W\times C}$, where $H$ and $W$ are spatial height and width and $C$ is the number of channels.

*1) Calculate the Correlation Matrix $\mathcal{C}$:* For the input two 3-D features $f_1$, $f_2 \in \mathbb{R}^{H\times W\times C}$, we reshape the features to $f_1^r$, $f_2^r \in \mathbb{R}^{n\times C}$, where $n = H\times W$. Then we compute pairwise channel correlations, obtaining a covariance matrix $\mathcal{C} \in \mathbb{R}^{C\times C}$. The calculation process is shown in the following equation:

$$\mathcal{C}_{ij} = \left(f_1^{ri} - \overline{f_1^{ri}}\right)\left(f_2^{rj} - \overline{f_2^{rj}}\right) \quad i, j = 1, 2, \ldots C \tag{9}$$

where $\mathcal{C}_{ij}$ represents the element of the $i$th row and $j$th column of the matrix $\mathcal{C}$, $f_1^{ri}$, $\overline{f_1^{ri}}$, $f_2^{rj}$, $\overline{f_2^{rj}}$ represent the $i$th channel of $f_1$, the average of the $i$th channel of $f_1$, the $j$th channel of $f_2$, and the average of the $j$th channel of $f_2$., respectively. The resulting covariance matrix has clear physical meaning, i.e., $\mathcal{C}_{ij}$ represents the correlation between the $i$th channel of $f_1$ and the $j$th channel of $f_2$. The information contained in matrix $C$ can help us analyze the statistical dependence between $f_1$ and $f_2$. By analyzing the statistical dependencies between features, we can help us explore effective ways to fuse features.

*2) Calculate Feature Fusion Mask $\alpha$ and $\beta$:* we sum the rows and columns of $\mathcal{C}$ separately, obtaining two column vectors $\text{row}_{\text{sum}}, \text{col}_{\text{sum}} \in \mathbb{R}^{C\times 1}$. The value of $\text{row}_{\text{sum}}^i(col_{\text{sum}}^i)$ represents the degree of correlation between $f_1^i(f_2^i)$ and $f_2(f_1)$.

The results of the cumulative sum vector can be divided into three types

$$\text{row}_{\text{sum}}^i\left(\text{col}_{\text{sumi}}^i\right)\begin{cases} > 0 \\ = 0 \\ < 0. \end{cases} \tag{10}$$

For the $\text{row}_{\text{sum}}$, if $\text{row}_{\text{sum}}^i > 0$, it means that the $f_1^i$ has a high relevance with $f_2$, which indicates that the information in $f_1^i$ can be similar to the information of many channels in $f_2$. Thus, the information $f_1^i$ contains is relatively redundant. In a general feature fusion method, the redundant information of the fusion feature will be automatically enhanced, which will reduce the discriminability of the features, resulting in a decrease in the accuracy of subsequent classification. Therefore, the redundant information needs to be suppressed in the subsequent fusion. If $\text{row}_{\text{sum}}^i > 0$ means that the channel has a low relevance with $f_2$. Thus the information it contains is relatively independent. Based on the previous analysis, it needs to be strengthened in the subsequent fusion. If $\text{row}_{\text{sum}}^i = 0$ means that the channel is neutral. In this case, the fusion process degenerates into traditional element-wise adding. The conclusion is the same for the $\text{col}_{\text{sum}}$.

In the attention model, the feature mask can effectively strengthen or inhibit the characteristic channels. Similarly, we generate feature masks based on the above feature correlation analysis. The generated mask can suppress or strengthen the channels to be fused according to the correlation. The equation to convert the $\text{row}_{\text{sum}}, \text{col}_{\text{sum}}$ to [0,1] is as follows:

$$\begin{aligned} f(x) &= 1 - \text{sigmoid}(x) \\ &= \frac{e^{-x}}{1+e^{-x}}. \end{aligned} \tag{11}$$

Different from the Sigmoid function, it is a negative feedback mapping, that is, the larger the input, the smaller the mapping value, vice versa. The mask will suppress feature channels with high redundancy and enhance feature channels with high independence. Therefore, we choose to use this function to generate the weights ($\alpha$ and $\beta$) for feature fusion

$$\begin{aligned} \alpha &= f(\text{row}_{\text{sum}}) = \frac{e^{-\text{row}_{\text{sum}}}}{1+e^{row_{\text{sum}}}} \\ \beta &= f(\text{col}_{\text{sum}}) = \frac{e^{-\text{col}_{\text{sum}}}}{1+e^{\text{col}_{\text{sum}}}}. \end{aligned} \tag{12}$$

*3) Obtain the Fusion Feature $f_3$:* We got the fusion feature $f_3$ by following equation:

$$f_3 = f_1' + f_2' = \alpha \cdot f_1 + \beta \cdot f_2. \tag{13}$$

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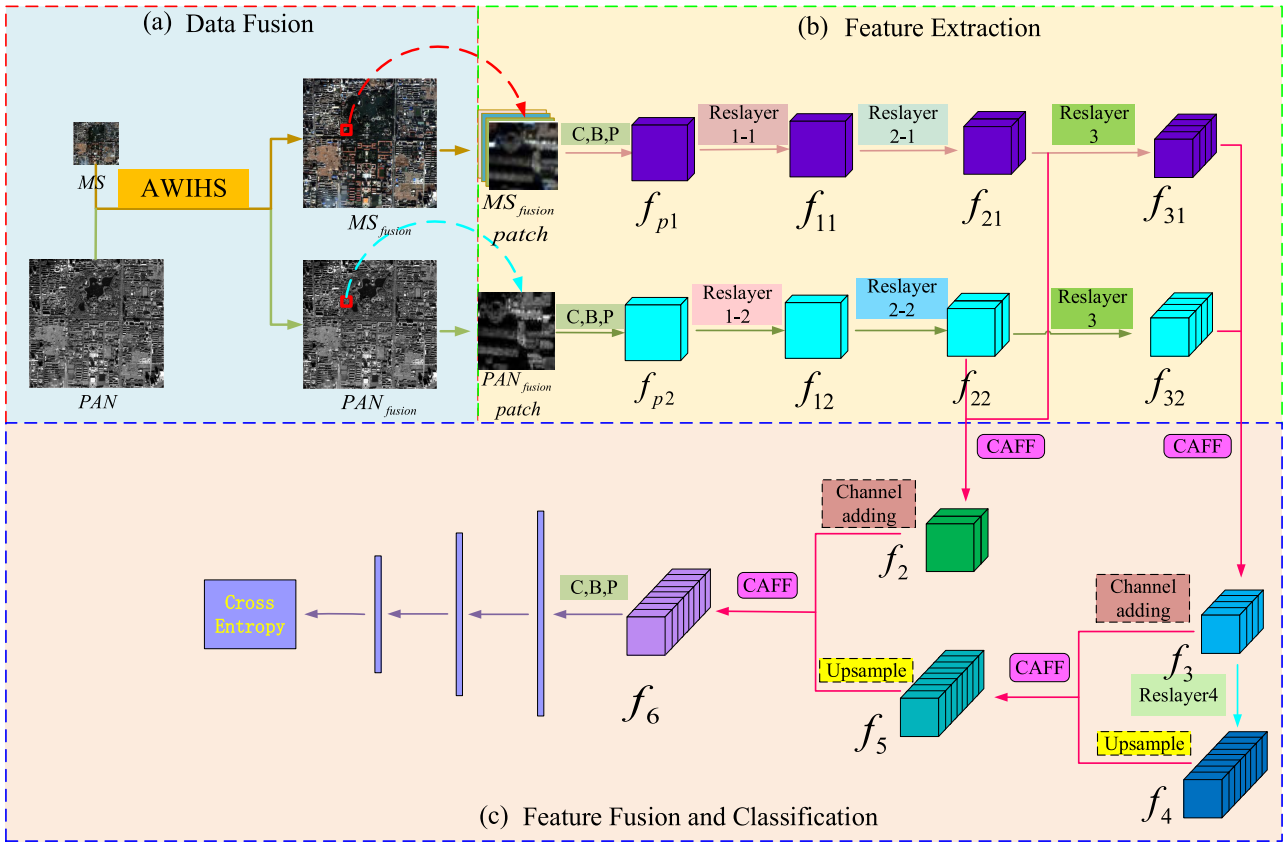



Fig. 3. Flowchart of AHF-Net. The network can be divided into three parts. (a) AWIHS data fusion. (b) Feature extraction net. (c) Feature fusion and classification. The bone network is dual-branch Resnet-18.

Through $\alpha$ and $\beta$, we can perform effective fusion based on the relative redundancy of $f_1$, $f_2$.

The PAN and MS images of the same scene are internally consistent but externally different. Meanwhile, they also have the problem of imbalance in the scale of the terrain that is common in remote sensing images. Therefore, we propose the AWIHS strategy and the CAFF module to help reduce data differences and perform effective multilevel feature fusion. Through the AWIHS strategy, the difference between PAN and MS is significantly reduced. We believe that the features' redundant parts belong to the shared background information from the perspective of correlation. The CAFF module can enhance their unique information and suppress redundant information to effectively alleviate the problem of reduced feature discrimination during feature fusion. Because the fusion is based on the second-order correlation of input features, the CAFF module can well inherit the dependencies of input features too. Therefore, our AHF-Net can improve the discrimination of fusion features.

### C. Construction of the AHF-Net

In this section, we introduce the AHF-Net in detail. As shown in Fig. 3, the entire network can be divided into three parts, where Fig. 3(a) is the AWIHS data fusion strategy, Fig. 3(b) is a backbone network for extracting features, and Fig. 3(c) is a fusion network for fusing multilevel features. The details are as follows:

*1) Data Fusion:* The AWIHS algorithm is used to reduce the difference between PAN and MS images. This module takes the original MS and PAN images as input, and outputs the fusion images $\text{MS}_{\text{fusion}}$ and $\text{PAN}_{\text{fusion}}$ after AWIHS data fusion algorithm.

*2) Feature Extraction:* As shown in Fig. 3(b), the inputs of the two branches are MS and PAN patches, and the outputs are $f_{31}$ and $f_{32}$, respectively. The first two Reslayers of the two branches have different subscripts, which indicates that their parameters are not shared. In contrast, the parameters of the third Reslayer are shared. We consider that the shallow features are the specific details of the image and the expression of the representation information. Even after the AWIHS data fusion, the differences in the details of the image still exist. So we use ResNet [33] blocks without parameters shared. With the deepening of the network, the extracted deep feature is a high-level semantic feature. These features should reflect the essential consistency of the image, so we use the ResNet blocks with parameters shared.

*3) Feature Fusion and Classification:* Due to the long imaging distance, the different types of terrain in PAN and MS images have large-scale differences. This inherent characteristic will decrease the accuracy of image classification. Since the deep network extracts features layer-by-layer, these features have an inherent pyramid shape [43]. Among them, the shallow features are rich in texture information, while the deep features have rich semantic information. Combining these

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different levels of features can help us improve the accuracy of classification [44].

The general method of feature fusion at different levels is element-wise adding or feature concatenating. In the two fusion methods, the weights of all feature channels are the same; that is, all feature channels' importance is the same. However, in actual situations, the features of different levels have different importance for the final classification task. Based on this, we integrated the CAFF module into FPN [43]. The corresponding channels are suppressed or enhanced according to the correlation between feature channels. The feature channels that are favorable for classification will be highlighted to improve the accuracy of classification.

The structure of the multilevel fusion network is shown in Fig. 3(c). Before constructing a multilayer feature fusion network, we first use CAFF to fuse features from two branches. Since the shallow features are the specific details of the image and the expression of the representation information, we use $f_{21}$ and $f_{22}$ as the first-level features for fusion. The fused feature $f_2$ can be expressed as follows:

$$f_2 = \alpha \cdot f_{21} + \beta \cdot f_{22} \tag{14}$$

where $\alpha$ and $\beta$ are the fusion coefficients. Then we combine the fusion features ($f_2$, $f_3$) with $f_4$ to form a three-level feature pyramid network. Before fusion, we use $1 \times 1$ convolution to add the channels of $f_2$ and $f_3$ to the number of $f_4$. Simultaneously, we use upsampling to increase the size of $f_4$ to $f_3$ and $f_5$ to $f_2$. Finally, CAFF is used to fuse features of different levels separately. After the fused feature pass through an average pooling layer and several fully connected layers, the class probability of the pair of patches is finally estimated. In this article, the cross-entropy error used as the ultimate loss function and defined as follows:

$$\mathrm{E} = -\frac{1}{n_b}\sum_{i=1}^{n_b}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{15}$$

where $n_b$ denotes the batch size, $y_i$ is the label for the $i$th input pair, while $\hat{y}_i$ is the class probability for the $i$th input pair.

## IV. Experimental Results

In this section, the proposed method will be evaluated on the data set of different areas, and we also compare our approach with several state-of-art algorithms. The experimental results and analysis will be given below.

### A. Data sets and Experimental Settings

*1) Data sets:* In this article, we use four sets of corresponding PAN and MS images in the experiment. The first three data sets [Fig. 4(a)–(c)] are obtained by the GaoFen I sensor. The images in Fig. 4(a) are all Level 1A data (pre-processed radiation correction image products): image data processed by data analysis, homogenization radiation correction, denoising, MTFC, CCD stitching, band registration, etc. and provide satellite direct-pass attitude and orbit data Production RPC file. It was acquired on August 29, 2015, in Xi'an, China. For the images in Fig. 4(a), the MS image consists of $4548 \times 4541 \times 4$ pixels with a spatial resolution of 8 m, while the PAN image consists of $18\,192 \times 18\,164$ pixels with a spatial resolution of 2 m. The data are divided into 12 categories: Water, Road1, Road2, Bareland, Lowvegetation, Farmland, Tree, Urban dense high-rise buildings, Urban sparse residential buildings, Suburban low-rise buildings, Blue roof buildings, and White roof buildings. Fig. 4(b) and (c) are all Level 1A data. They were acquired in Hohhot China on May 23, 2015, and Nanjing China on April 21, 2015. For the images in Fig. 4(b), the MS image consists of $2001 \times 2101 \times 4$ pixels with a spatial resolution of 4 m, while the PAN image consists of $8004 \times 8404$ pixels with a spatial resolution of 1 m. The data are divided into eleven categories: Water, Road1, Bareland, Vegetation, Urban dense high-rise buildings, Urban sparse residential buildings, Suburban low-rise buildings, Farmland, White roof buildings, Red roof buildings, and Blue roof buildings. For the images in Fig. 4(c), the MS image consists of $2000 \times 2500 \times 4$ pixels with a spatial resolution of 4 m, while the PAN image consists of $8000 \times 10\,000$ pixels with a spatial resolution of 1 m. The data is divided into eleven categories: Water, Road1, Bareland, Vegetation, Red roof buildings, Blue roof buildings, Urban dense high-rise buildings, White roof buildings, Urban sparse residential buildings, Road2, and Lowvegetation.

The last data set [Fig. 4(d)] is obtained by the QuickBird sensor. These images were acquired in Xian, China, on May 30, 2008. For the images in Fig. 4(d), the MS image consists of $800 \times 830 \times 4$ pixels with a spatial resolution of 2.44 m, while the PAN image consists of $3200 \times 3320$ pixels with a spatial resolution of 0.61 m. The data is divided into seven categories: Building, Flat-land, Road, Shadow, Soil, Tree, and Water.

The experiments in this article are running on Workstation with RTX2080Ti 12 GB GPU and 128 GB RAM under Ubuntu 16.04 LTS. The proposed network is trained on PyTorch.

*2) Experimental Settings:* For evaluating the classification performance, the metrics such as overall accuracy (OA), average accuracy (AA), and kappa statistic (Kappa) are adopted to make a quantitative analysis. The detailed network parameters of the proposed AHF-Net are shown in Table I. In this article, since the terrain of the four data sets is very similar, we use a unified network parameters.

In the training of the networks, we randomly select 2% labeled data from each class as training samples. In order to alleviate the impact of sample imbalance, we set a sampling threshold of 1500. When the sample size of 2% of a specific category is less than 1500, we increase this category's sample size to 1500. After selecting the training samples, the intersection-over-union (IOU) ratio is used as the criterion for choosing the test set. Specifically, pixels (IOU ratio of the training samples greater than 1/3) will not be included in the test set. The initial learning rate is 0.001, the weight decay is 0.0005, the batch size is 64, and the epoch is 30. To guarantee enough variability of the proposed method, we ran codes ten

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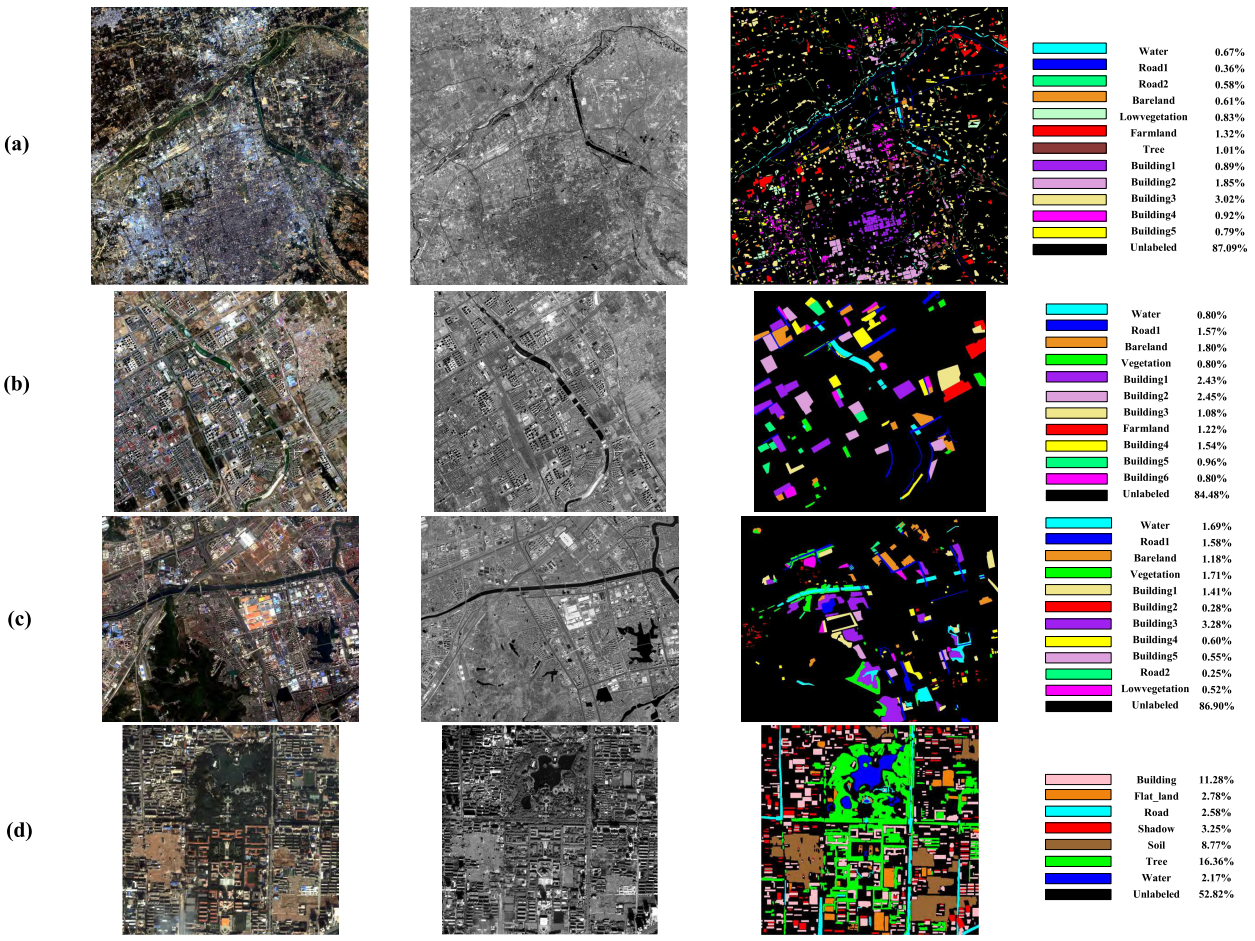



Fig. 4. First column: MS image. Second column: PAN images (The PAN image is reduced to the same size as their MS image for a more convenient display). Third column: Ground truth image. Last column: The class labels corresponding to the ground-truth image. (a) Xi'an Level 1A image set. (b) Hohhot Level 1A image set. (c) Nanjing Level 1A image set. (d) Xi'an Ubarn image set.

times on different groups of random training samples and took the average results and STD as the final result for each metric.

### *B. Comparison and Analysis of Hyperparameters*

In this section, we do a detailed comparison and analysis of hyperparameters: patch size of the network and the reslayer number shown in Fig. 3. Here, except for the selected hyperparameters, each set of data is trained and tested using an AHF network with the same other parameters. We used the fixed variable method to carry out the experiment.

A quantitative comparison on the different data sets is shown in Tables II and III. It can be seen that the classification performance is the highest when the patch size is 80, and the reslayer is 4. For the patch size, we all know that the patch's size affects the amount of neighborhood information of the input image patch. When the patch size is 64, the classification performance is relatively low due to insufficient neighborhood information. When the patch size is 96, due to the redundancy of domain information, the algorithm is easy to fall into the local optimal, so the classification performance will also decline. Therefore, We chose 80 as patch size instead of 64 or 96. For the reslayer number, when the network layers are not deep enough, the extracted features are not robust enough, so the classification performance is not high. When the number of layers exceeds a certain depth, the network is prone to overfitting, which reduces the classification performance. We chose four as the number of reslayers instead of 3 or 5. In addition, in parameter selection, we also found that the STD of (patch size 80, reslayer 4) AHF net algorithm is also smaller $(-0.2, 0.2)$. This result shows that the stability of the network is the highest under this parameter.

### *C. Performance of the Proposed AWIHS Strategy and CAFF*

In this section, taking Xi'an Urban image set as an example, we do a detailed comparison and analysis of the AWIHS strategy and CAFF module.

*1) Validation of the AWIHS Strategy:* In this part, we use AIHS [13] and IAIHS [15] algorithms to compare with the proposed AWIHS algorithm. Our goal is to verify whether AWIHS can reduce the data difference between PAN and MS and alleviate spectral distortion. For the experiment's fairness, the three algorithms use the same edge detection operator and intensity component ($I_c$), except for the fusion coefficient. In the subsequent classification, the three methods also use AHF-Net with the same parameters and iteration times. In order to better verify the performance of AWIHS, besides the classification accuracy metrics (OA, AA, Kappa), we also introduce two metrics: the relative average spectral error (RASE) commonly used to evaluate image differences;

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TABLE I

HYPERPARAMETERS OF EACH NETWORK LAYER

| Patch | PAN(MS) | | | | |
|---|---|---|---|---|---|
| Types | Input | Filter Size | Stride | Pooling | Output |
| Preprocess | $S \times S, 1$ ($S \times S, 4$) | $3 \times 3, 16$ | 1 | Yes | $S/2 \times S/2, 16$ |
| $Layer_1$ $(unshared)$ | $S/2 \times S/2, 16$ | $\begin{bmatrix} 3 \times 3, 16 \\ 3 \times 3, 16 \end{bmatrix} \times 2$ | 1 | No | $S/2 \times S/2, 16$ |
| $Layer_2$ $(unshared)$ | $S/2 \times S/2, 16$ | $\begin{bmatrix} 3 \times 3, 32 \\ 3 \times 3, 32 \end{bmatrix} \times 2$ | 2 | Yes | $S/4 \times S/4, 32$ |
| $Branch\ Fusion\ Layer_1$ | $S/4 \times S/4, 32$ | - | | | $S/4 \times S/4, 32$ |
| $Channel\ Add\ Layer_1$ | $S/4 \times S/4, 32$ | $1 \times 1, 128$ | 1 | No | $S/4 \times S/4, 128$ |
| $Layer_3$ $(shared)$ | $S/4 \times S/4, 32$ | $\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$ | 2 | Yes | $S/8 \times S/8, 64$ |
| $Branch\ Fusion\ Layer_2$ | $S/8 \times S/8, 64$ | - | | | $S/8 \times S/8, 64$ |
| $Channel\ Add\ Layer_2$ | $S/8 \times S/8, 64$ | $1 \times 1, 128$ | 1 | No | $S/8 \times S/8, 128$ |
| $Layer_4$ $(fusion)$ | $S/8 \times S/8, 64$ | $\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$ | 2 | Yes | $S/16 \times S/16, 128$ |
| $Channel\ Add\ Layer_3$ | $S/16 \times S/16, 128$ | $1 \times 1, 128$ | 1 | No | $S/16 \times S/16, 128$ |
| $Multilevel\ Fusion\ Layer_1$ | $S/8 \times S/8, 128$ $S/16 \times S/16, 128$ | - | | | $S/8 \times S/8, 128$ |
| $Multilevel\ Fusion\ Layer_2$ | $S/4 \times S/4, 128$ $S/8 \times S/8, 128$ | - | | | $S/4 \times S/4, 128$ |
| Smooth | $S/4 \times S/4, 128$ | $1 \times 1, 128$ | 1 | Yes | $S/8 \times S/8128$ |
| FC | $S/8 \times S/8128$ | - | | | class |

TABLE II

EXPERIMENTAL RESULTS OF HYPERPARAMETER PATCH SIZE

| Data Sets | Xi'an Level 1A | | | Hohhot Level 1A | | | Nanjing Level 1A | | | Xi'an Urban | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Patch Size | 64 | 80 | 96 | 64 | 80 | 96 | 64 | 80 | 96 | 64 | 80 | 96 |
| OA(%) | 95.94±0.2 | **96.59**±0.1 | 94.46±0.1 | 93.97±0.3 | **96.82**±0.1 | 94.44±0.2 | 95.26±0.2 | **96.13**±0.1 | 95.85±0.2 | 98.02±0.1 | **99.28**±0.1 | 98.43±0.1 |
| AA(%) | 95.24±0.3 | **96.18**±0.2 | 93.69±0.2 | 94.20±0.3 | **96.64**±0.1 | 94.62±0.2 | 95.26±0.1 | **96.46**±0.2 | 96.14±0.2 | 97.76±0.2 | **99.20**±0.2 | 98.33±0.1 |
| Kappa(%) | 95.39±0.3 | **95.91**±0.2 | 93.72±0.4 | 93.24±0.3 | **96.27**±0.1 | 93.78±0.1 | 94.50±0.2 | **95.51**±0.1 | 95.35±0.2 | 97.39±0.1 | **99.09**±0.1 | 97.98±0.1 |

TABLE III

EXPERIMENTAL RESULTS OF HYPERPARAMETER RESLAYER NUMBER

| Data Sets | Xi'an Level 1A | | | Hohhot Level 1A | | | Nanjing Level 1A | | | Xi'an Urban | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Res Number | 3 | 4 | 5 | 3 | 4 | 5 | 3 | 4 | 5 | 3 | 4 | 5 |
| OA(%) | 95.79±0.3 | **96.59**±0.1 | 94.94±0.2 | 94.20±0.6 | **96.82**±0.1 | 95.41±0.5 | 95.02±0.3 | **96.13**±0.1 | 95.20±0.4 | 97.85±0.1 | **99.28**±0.1 | 97.34±0.5 |
| AA(%) | 95.01±0.3 | **96.18**±0.2 | 93.76±0.5 | 94.55±0.6 | **96.64**±0.1 | 95.54±0.5 | 95.49±0.3 | **96.46**±0.2 | 95.46±0.3 | 97.56±0.2 | **99.20**±0.2 | 96.93±0.1 |
| Kappa(%) | 95.23±0.3 | **95.91**±0.2 | 93.13±0.4 | 93.38±0.6 | **96.27**±0.1 | 94.98±0.6 | 94.23±0.3 | **95.51**±0.1 | 94.27±0.5 | 97.23±0.1 | **99.09**±0.1 | 96.57±0.1 |

the relative global adimensional synthesis error (RGASE) that evaluates the spectral distortion of pansharpening algorithm. The calculation method of the two is as follows:

$$\mathrm{RASE} = \frac{100}{\overline{\mathrm{MS}_{\mathrm{fusion}}}} \sqrt{\frac{1}{n}\sum_{i=1}^{n} \mathrm{RMSE}^2\left(\mathrm{MS}^i_{\mathrm{fusion}}, \mathrm{PAN}_{\mathrm{fusion}}\right)}$$

$$\mathrm{RGASE} = 100\frac{h}{l} \sqrt{\frac{1}{n}\sum_{i=1}^{n} \frac{\mathrm{RMSE}^2\left(\mathrm{MS}^i_{\mathrm{fusion}}, \mathrm{PAN}_{\mathrm{fusion}}\right)}{\overline{\mathrm{MS}^i_{\mathrm{fusion}}}^2}} \quad (16)$$

where RMSE refers to the root mean square error, $\overline{\mathrm{MS}_{\mathrm{fusion}}}$, $\overline{\mathrm{MS}^i_{\mathrm{fusion}}}$, $h$ and $l$ refer to the $\mathrm{MS}_{\mathrm{fusion}}$ overall mean, the $\mathrm{MS}_{\mathrm{fusion}}$ $i$th band mean, the spatial resolution of PAN and the original MS, respectively.

Quantitative comparison and qualitative comparison of the Xi'an Urban image data set are shown in Table IV and Fig. 5, respectively. From Fig. 5, it can be seen that the color of Fig. 5(d) is the closest to that of Fig. 5(a), and its spatial details are also improved. Take the content in the red frame in the figure as an example. The AIHS [Fig. 5(b)], IAIHS [Fig. 5(c)] and AWIHS [Fig. 5(d)] has all improved the details of the MS image texture. Specifically, the texture of trees becomes more detailed, and the edges of small paths and buildings in the picture become clearer. However, in Fig. 5(b) and (c), there are varying degrees of white edges on the edges of the house (as shown in the yellow circle). The tree color in the picture also shows obvious deviation (Fig. 5(b) is yellow-green, while Fig. 5(c) is light green, as shown in the red circle). These are the spectral distortion caused by the algorithms. In comparison, our algorithm [Fig. 5(d)] significantly reduces the distortion of the spectrum, making the color of the objects in the picture closer to the original image. Table IV shows that AWIHS achieves the highest classification results and the smallest value of different metrics (RASE: 57, RGASE: 1.24). The classification results indirectly prove the effectiveness of AWIHS. Therefore, our algorithm can effectively reduce the difference between MS and pan data and alleviate spectral distortion.

*2) Validation of the CAFF Module:* In this section, the conventional fusion strategy feature element-wise adding and feature concatenating fusion strategies are used to compare with the proposed CAFF module. The network used by

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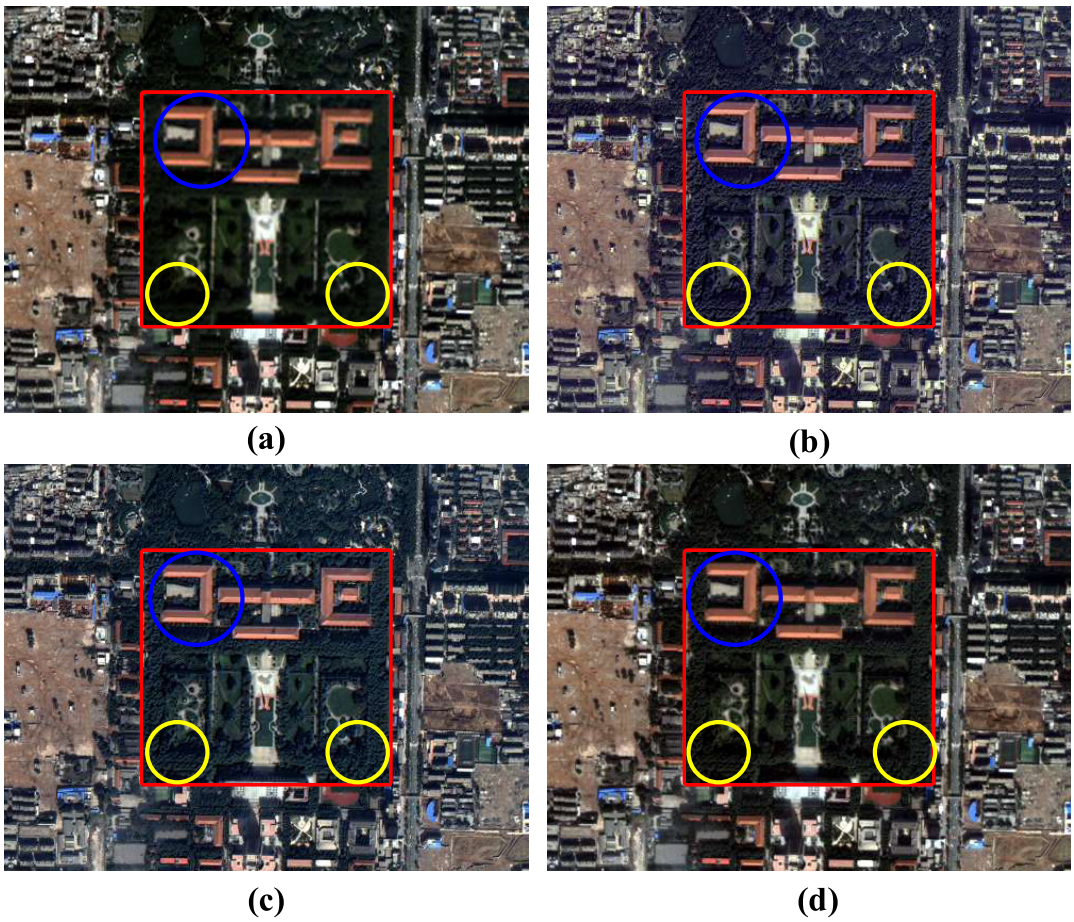



Fig. 5. Pansharpening results. (a) Original MS image. (b) AIHS. (c) IAIHS. (d) AWIHS.

TABLE IV
QUANTITATIVE COMPARISON OVER DIFFERENT PANSHARPENING ALGORITHMS ON THE XI'AN URBAN IMAGE DATA

| Feature Fusion Methods | AIHS | AWIHS | IAIHS |
|---|---|---|---|
| $c_1(Building)(\%)$ | 97.50±0.6 | **98.18**±0.2 | 97.76±0.4 |
| $c_2(Flat_land)(\%)$ | 95.92±0.2 | **99.92**±0.3 | 98.27±0.6 |
| $c_3(Road)(\%)$ | 95.29±0.3 | **99.21**±0.8 | 96.09±0.2 |
| $c_4(Shadow)(\%)$ | 94.78±0.2 | **99.48**±0.06 | 96.44±0.6 |
| $c_5(Soil)(\%)$ | 98.73±0.6 | **99.57**±0.4 | 99.32±0.2 |
| $c_6(Tree)(\%)$ | 97.78±0.5 | **99.13**±0.2 | 98.06±0.3 |
| $c_7(water)(\%)$ | 99.18±0.5 | **99.66**±0.3 | 99.16±0.5 |
| OA(%) | 97.46±0.5 | **99.28**±0.1 | 98.04±0.1 |
| AA(%) | 97.03±0.7 | **99.20**±0.2 | 97.87±0.2 |
| Kappa(%) | 96.72±0.6 | **99.09**±0.1 | 97.48±0.2 |
| RASE | 71 | **57** | 64 |
| RGASE | 6.07 | **1.24** | 4.09 |

these fusion strategies is our AHF-Net, and network-related hyperparameters are also the same for fair.

The experimental result is shown in Table V. It can be seen that our CAFF module obtains the highest classification results (OA, AA, and Kappa). Under the premise of equal fusion weight, the classification results of accumulation and cascade are not different. The advantage of element-wise adding is that the parameters are relatively small, while feature concatenating can completely keep all the feature information. For the features to be fused, the general equal-weight fusion strategy does not suppress and strengthen the specific feature channel, resulting in the fused features' not inheriting the input features' discriminative information well. Hence, the classification accuracy is relatively low. Our CAFF module can effectively alleviate the above problems, so our module can improve classification performance.

TABLE V
QUANTITATIVE COMPARISON OVER DIFFERENT FEATURE FUSION STRATEGIES ON XI'AN URBAN IMAGE DATA

| Fusion Methods | Element-wise Adding | CAFF | Concatenating |
|---|---|---|---|
| $c_1(Building)(\%)$ | 97.35±0.2 | **98.18**±0.2 | 97.61±0.4 |
| $c_2(Flat_land)(\%)$ | 97.96±0.9 | **99.92**±0.3 | 98.17±0.9 |
| $c_3(Road)(\%)$ | 96.77±0.6 | **99.21**±0.8 | 97.65±0.7 |
| $c_4(Shadow)(\%)$ | 95.42±0.5 | **99.48**±0.06 | 95.59±1.1 |
| $c_5(Soil)(\%)$ | 99.35±0.1 | **99.57**±0.4 | 99.36±0.1 |
| $c_6(Tree)(\%)$ | 97.91±0.3 | **99.13**±0.2 | 98.99±0.4 |
| $c_7(Water)(\%)$ | 99.17±0.2 | **99.66**±0.3 | 99.13±0.3 |
| OA(%) | 97.84±0.1 | **99.28**±0.1 | 98.11±0.1 |
| AA(%) | 97.71±0.1 | **99.20**±0.2 | 98.33±0.2 |
| Kappa(%) | 97.22±0.1 | **99.09**±0.1 | 98.01±0.1 |

### *D. Ablation Experiment and Comparison Algorithm*

In this section, to verify the effectiveness of the algorithm, we conduct the corresponding baseline, and ablation experiments, namely $\text{Net}_{\text{pansharp}}$, FPN [43]+Dual-branch Res18

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(FDBR18), AWIHS+FDBR18, and CAFF+FDBR18. Among them, $\text{Net}_{\text{pansharp}}$ and FDBR18 are used as baseline methods.

The two inputs of $\text{Net}_{\text{pansharp}}$ are pansharpening MS image and PAN image. The pansharpening algorithm it applies is AIHS. The structure of the network model adopts the structure of FuseNet [25]. FDBR18 uses the original PAN and MS images as input. The first three layers of FDBR18 are two branches used to extract the features of PAN and MS. The fourth layer is the fusion flayer, and the network becomes a single branch. The last three layers of the network constitute the pyramid network. FPN is followed by several fully connected layers. In order to be fair, the parameters (learning rate, sampling rate, patch size, and so on) of the FDBR18 network are the same as AHF-Net.

Besides, we compare the proposed AHF-Net with three state-of-the-art methods ACO-SS + DBFA-Net [27], ReuseNet [26] and SML-CNN [23]. AHF net consists of the following three parts: AWIHS, CAFF, and FDBR18. For all networks, we follow the experimental setup (10% label data while our AHF-net 2% of that) in their respective papers to achieve their best results. The comparison and analysis of the experimental results of different methods are as follows. Moreover, we give the test time to compare the efficiency of each method.

Taking Xi'an data as an example, we compared our AHF-Net with other methods. The specific analysis is as follows:

*1) $\text{Net}_{\text{pansharp}}$ and FDBR18 Vs AWIHS+FDBR18:* For Xi'an Level 1A Images, The corresponding quantitative and qualitative results are shown in Table VI and Fig. 6 (Due to space constraints, the AWIHS + FDBR18 and CAFF + FDBR18 are not shown in Fig. 6, and neither do the experimental results of the following data sets). The AWIHS obtain higher results than $\text{Net}_{\text{pansahrp}}$ and FDBR18. AWIHS improves the accuracy of most categories, especially c4 (Bareland) and c5 (Lowvegetation) significantly improved the accuracy from 84.25% ($\text{Net}_{\text{pansharp}}$),79.43% (FDBR18) to 90.61% and from 74.40% ($\text{Net}_{\text{pansharp}}$), 79.42% (FDBR18) to 91.20%. As can be seen from the ground truth map, the pixels of c4 and c5 are widely distributed and contains objects of multiple shapes, and the spectral information is similar to c6 (Farmland). In the imaging process of MS images, the spectra of different terrain may be similar. Therefore, we need to use PAN with a high spatial resolution to help distinguish these similar terrains. However, although the PAN image and the MS image are essentially the same, there are still external differences, so we need to consider reducing the difference between the MS and PAN image instead of directly using these two images for classification. It is noted that due to the limited performance of the FDBR18 Net, AWIHS strategy does not raise too much accuracy, some categories [c1 (Water), c7 (Tree), c10 (Building3)] obtain low accuracy. Compared with the other results of Table VI, the overall OA, AA, Kappa of the above three methods is not very high.

*2) $\text{Net}_{\text{pansharp}}$ and FDBR18 Vs CAFF+FDBR18:* Compared with the $\text{Net}_{\text{pansharp}}$ and FDBR18, the CAFF+FDBR18 Net obtains higher results. Our CAFF+FDBR18 Net can improve the accuracy of most categories too. Due to imaging reasons, remote sensing terrain usually has significant scale differences. Different scales of terrain generally have different preferences for feature channels. Therefore, a fusion strategy that can reflect the importance of feature channels is needed. Our CAFF module can alleviate this problem so that it can improve classification accuracy. However, the accuracies of some categories [e.g., c1 (Water), c7(Tree), and c10 (Building3)] have not been improved much. This is probably because these categories' spectral characteristics are quite different from other categories, and the spatial scale difference is small. Therefore, these categories in MS and pan images have little difference, so they do not rely on data difference reduction and feature fusion strategy.

*3) AWIHS+FDBR18, CAFF+FDBR18 Vs AHF-Net:* Table VI shows that the classification results of most categories of AHF-Net are higher than AWIHS+FDBR18 and CAFF+FDBR18 and the OA (OA, AA, Kappa) of AHF-Net is also the highest. This result shows that AWIHS and CAFF are not in conflict. AHF-Net inherits the advantages of the AWIHS and CAFF so that the classification accuracy can be further improved. The accuracy of c3 (Road2) of AHF-Net in the table is lower than that of CAFF+FDBR18. We analyze that it may be due to the close characteristics of c3 (Road2) and c2 (Road1), and AWIHS also has an inevitable spectral distortion, so AHF-Net's classification accuracy c3 is reduced. In addition, due to the reasonable design of the network model, the STD of the CAFF+FDBR18 (−0.2–0.2) and AHF-Net (−0.2–0.2) are lower than FDBR18 (−2.6–2.6) and AWIHS+FDBR18 (−0.9–0.9).

*4) AHF-Net Versus ReuseNet, SML-CNN and ACO-SS+DBAF-Net:* Moreover, our AHF-Net is also better than the results of three state-of-the-art methods. The structure of ReuseNet recycles FuseNet. Different from the patch classification method, where all the information of a patch serves only the central pixel, the full convolution method adopted by ReuseNet needs to determine the category of all the pixels in the patch. In addition, because the number of convolution layers is relatively small, the features extracted by ReuseNet may not be robust enough. Therefore, the accuracy is slightly lower than that of ACO-SS+DBAF-Net. However, due to the full convolution classification method, the classification efficiency of ReuseNet is very high. The test time is the shortest in different data sets.

SML-CNN first super-pixel segments the MS image. It then uses six local regions of super-pixels (four corner regions, one original region, and one central region) as input, and designed six CNNs for feature extraction. Compared with ReuseNet, it obtained lower results, in the category with fewer training samples, its accuracy is very low (e.g., c2 (Road1): 0.36%, c4 (Bareland): 0.61%, c7 (Building1): 0.89 %, c12 (Building5): 0.79%). After super-pixel segmentation, there are fewer labeled super-pixel samples, so in fact, SML-CNN uses more training data (25%). This makes the network easy to fall into local optimum, so its accuracy is lower than ACO-SS+DBAF-Net.

ACO-SS+DBAF-Net uses an adaptive strategy to select the neighborhood information of samples and combines spatial and channel attention mechanisms to extract and fuse pan and

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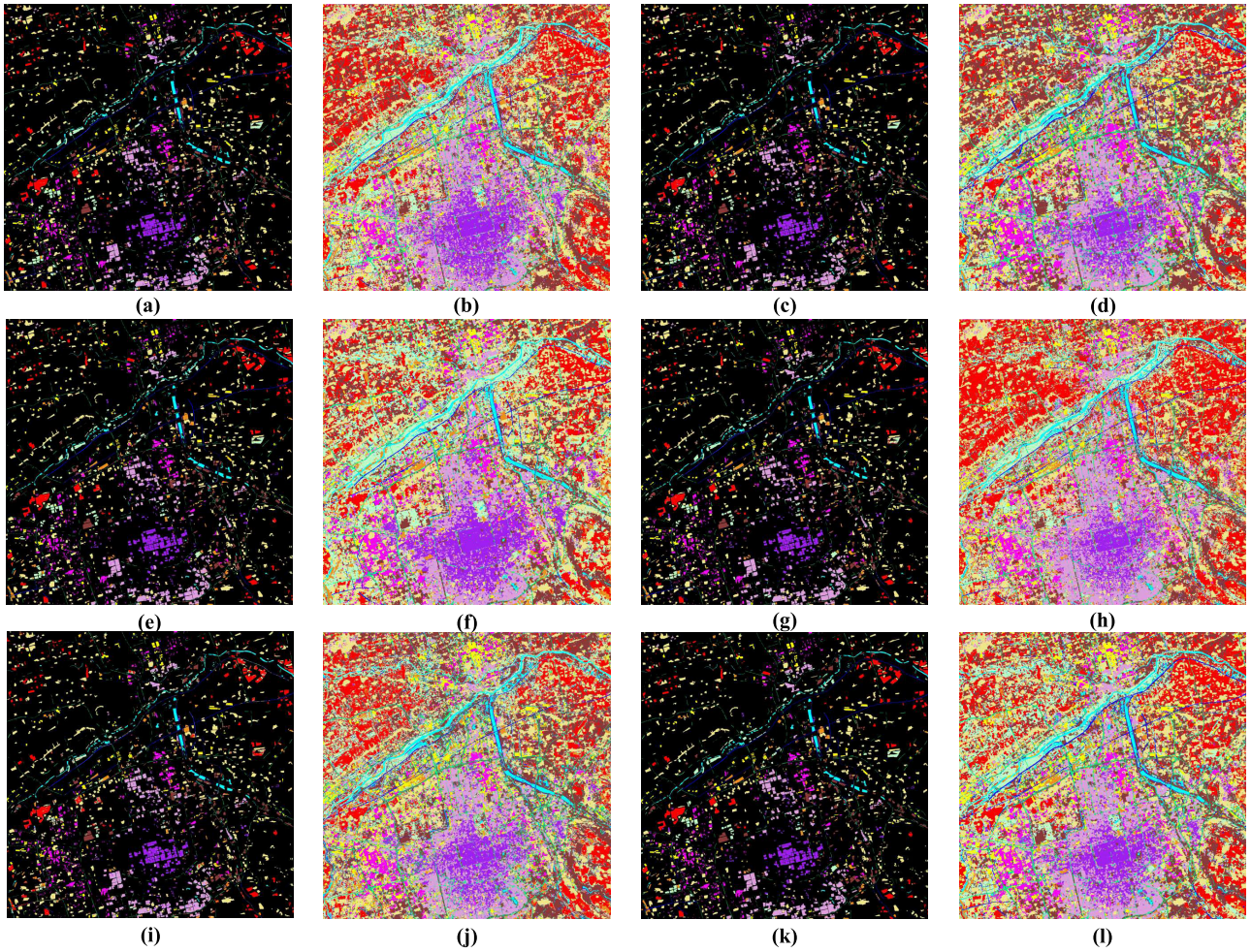



Fig. 6. Classification maps of different methods on the Xi'an Level 1A Images, each method has two classification maps: one is image with ground truth, while the other is overall image. (a) and (b) $\text{Net}_{\text{pansharp}}$. (c) and (d) FDBR18. (e) and (f) ACO-SS+DBAF-Net. (g) and (h) ReuseNet. (i) and (j) SML-CNN. (k) and (l) AHF-Net.

TABLE VI
QUANTITATIVE CLASSIFICATION RESULTS OF XI'AN LEVEL 1A IMAGES

| Methods | Net $_{pansharp}$ | FDBR18 | AWIHS+ FDBR18 | CAFF+ FDBR18 | ACO-SS+ DBFA-Net | ReuseNet | SML-CNN | AHF-Net |
|---|---|---|---|---|---|---|---|---|
| $c_1(Water)(\%)$ | 98.46 | 98.38±0.6 | 97.87±0.2 | 98.94±0.2 | 98.78 | 97.94 | 97.90 | **99.02**±0.1 |
| $c_2(Road1)(\%)$ | 92.79 | 90.88±1.4 | 95.12±1.0 | 95.01±0.6 | 98.41 | 93.13 | 81.5 | **98.54**±0.3 |
| $c_3(Road2)(\%)$ | 97.95 | 91.94±2.6 | 94.97±0.9 | 95.32±0.1 | 92.92 | **97.86** | 86.68 | 93.12±0.6 |
| $c_4(Bareland)(\%)$ | 84.25 | 79.43±3.7 | 90.61±1.6 | 92.17±1.4 | 92.04 | 91.20 | 87.89 | **96.07**±0.7 |
| $c_5(Lowvegetation)(\%)$ | 74.40 | 79.42±1.8 | 91.2±2.9 | 91.04±1.3 | 85.31 | 83.86 | 83.34 | **94.49**±0.5 |
| $c_6(Farmland)(\%)$ | 88.36 | 95.71±2.2 | 97.78±0.6 | 97.96±0.3 | 99.08 | 92.95 | 93.66 | **99.09**±0.2 |
| $c_7(Tree)(\%)$ | 88.70 | 91.63±3.1 | 93.05±1.7 | 95.73±0.4 | **97.85** | 93.77 | 94.60 | 95.93±0.6 |
| $c_8(Building1)(\%)$ | 83.71 | 92.08±3.1 | 92.64±2.6 | 92.54±0.8 | 94.52 | 93.98 | 90.42 | **97.41**±0.5 |
| $c_9(Building2)(\%)$ | 81.61 | 92.81±1.4 | 93.99±1.6 | 95.10±0.4 | **98.61** | 91.07 | 92.94 | 97.89±0.2 |
| $c_{10}(Building3)(\%)$ | 95.61 | 97.66±2.6 | 96.13±0.3 | 97.26±0.1 | 95.76 | 97.72 | 95.96 | **97.96**±0.1 |
| $c_{11}(Building4)(\%)$ | 96.54 | 96.66±1.4 | 96.26±0.4 | 98.06±0.2 | 96.84 | 97.47 | 98.10 | **98.14**±0.1 |
| $c_{12}(Building5)(\%)$ | 90.04 | 92.18±3.6 | 92.07±1.3 | 95.63±0.7 | **98.12** | 94.48 | 92.95 | 96.38±0.3 |
| OA(%) | 91.37 | 92.18±2.6 | 94.02±0.6 | 96.24±0.1 | 95.91 | 94.41 | 92.88 | **96.59**±0.1 |
| AA(%) | 89.37 | 91.57±1.2 | 93.98±0.9 | 95.57±0.2 | 95.69 | 93.89 | 91.34 | **96.18**±0.2 |
| Kappa(%) | 90.20 | 91.35±1.7 | 93.57±0.5 | 95.73±0.2 | 94.70 | 93.75 | 91.92 | **95.91**±0.2 |
| Test Time(s) | **132.31** | 597.64±18.4 | 714.31±57.8 | 849.16±66.5 | 1125.78 | 186.54 | 702.96 | 1252.16±64.3 |

MS image features. Therefore, the classification accuracy of most categories of ACO-SS+DBAF-Net is higher than that of ReuseNet and SML-CNN. However, due to the external differences between the MS and PAN maps, the differences in the features extracted by ACO-SS+DBAF-Net are also large. Therefore, the classification performance of the network

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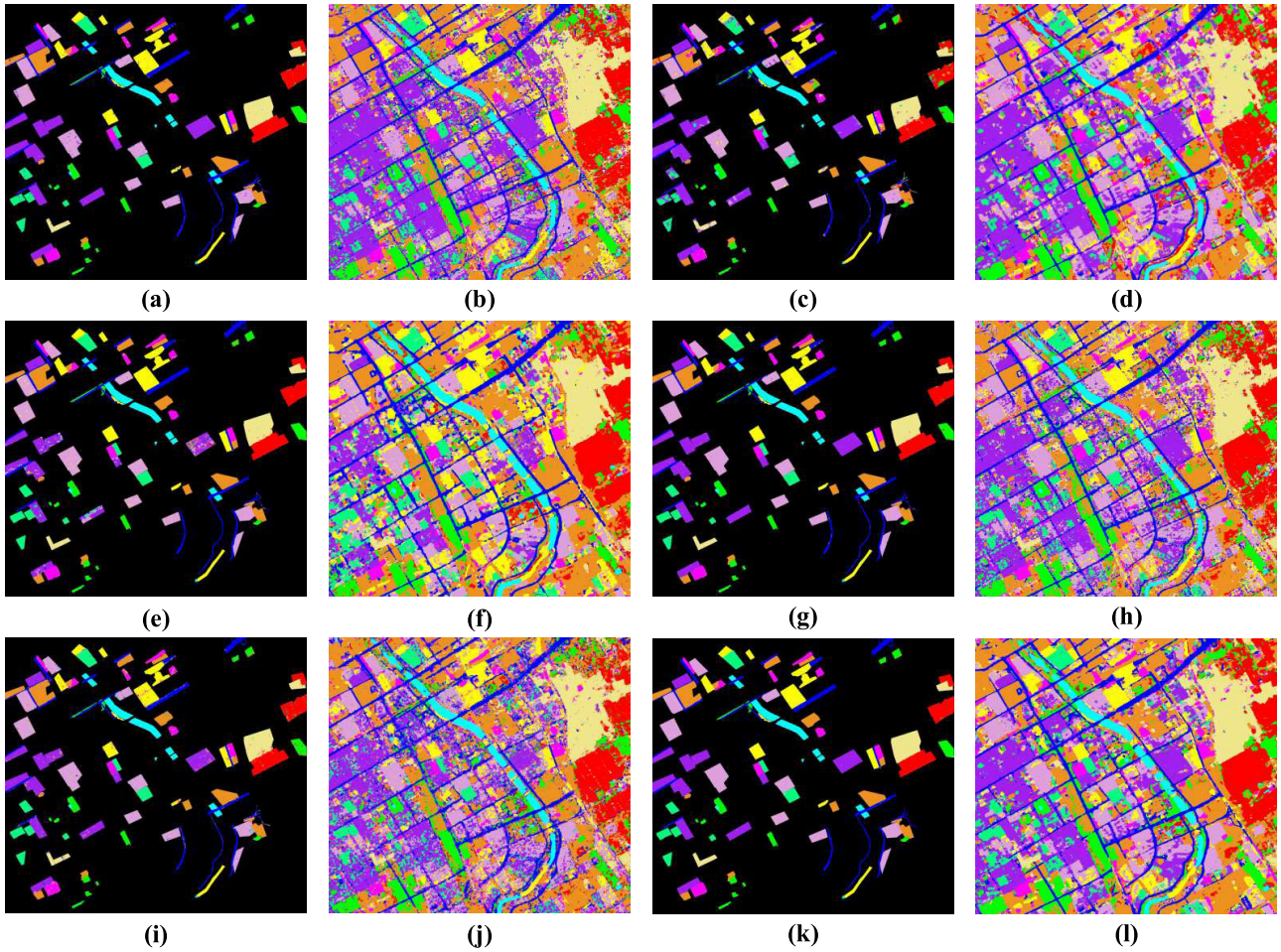



Fig. 7. Classification maps of different methods on the Hohhot Level 1A Images, each method has two classification maps: one is image with ground truth, while the other is overall image. (a) and (b) $\text{Net}_{\text{pansharp}}$. (c) and (d) FDBR18. (e) and (f) ACO-SS+DBAF-Net. (g) and (h) ReuseNet. (i) and (j) SML-CNN. (k) and (l) AHF-Net.

TABLE VII
QUANTITATIVE CLASSIFICATION RESULTS OF HOHHOT LEVEL 1A IMAGES

| Methods | Net $_{pansharp}$ | FDBR18 | AWIHS+ FDBR18 | CAFF+ FDBR18 | ACO-SS+ DBAF-Net | ReuseNet | SML-CNN | AHF-Net |
|---|---|---|---|---|---|---|---|---|
| $c_1(Water)(\%)$ | 98.70 | 98.32±1.1 | 99.04±0.2 | 99.28 | 99.21 | 99.20 | 99.18 | **99.22**±0.1 |
| $c_2(Road1)(\%)$ | 95.62 | 85.93±4.1 | 92.59±1.3 | 93.38±0.6 | 92.24 | 95.85 | 91.04 | **94.35**±0.3 |
| $c_3(Bareland)(\%)$ | 95.86 | 87.23±2.9 | 93.92±1.2 | 95.17±0.5 | 94.04 | 96.53 | 93.14 | **96.25**±0.3 |
| $c_4(Vegetation)(\%)$ | 97.16 | 89.38±2.1 | 93.32±1.5 | 94.71±1.0 | **97.60** | 95.12 | 91.59 | 96.81±0.3 |
| $c_5(Building1)(\%)$ | 96.18 | 85.38±3.8 | 93.65±2.4 | 93.64±0.5 | **99.97** | 95.64 | 92.74 | 98.52±0.4 |
| $c_6(Building2)(\%)$ | 94.29 | 88.14±3.5 | 94.42±1.0 | 94.28±0.7 | 91.14 | 97.17 | 94.72 | **97.23**±0.2 |
| $c_7(Building3)(\%)$ | 75.67 | 90.62±1.5 | 97.43±0.9 | 96.71±0.8 | 96.11 | 98.03 | 91.37 | **98.10**±0.3 |
| $c_8(Farmland)(\%)$ | 92.19 | 88.69±1.6 | 95.31±1.4 | 96.50±0.9 | 96.42 | 96.64 | 95.48 | **97.83**±0.3 |
| $c_9(Building4)(\%)$ | 72.34 | 84.98±1.2 | 94.35±1.8 | 94.70±0.8 | 92.60 | 96.15 | 93.41 | **96.22**±0.4 |
| $c_{10}(Building5)(\%)$ | 95.10 | 91.62±1.6 | 93.82±1.6 | 93.70±1.0 | 92.27 | 92.47 | 92.99 | **96.15**±0.3 |
| $c_{11}(Building6)(\%)$ | 95.19 | 86.55±3.7 | 92.88±0.1 | 93.28±0.8 | **97.51** | 95.04 | 88.82 | 95.69±0.3 |
| OA(%) | 91.15 | 89.14±3.1 | 94.37±0.8 | 94.76±0.2 | 94.80 | 95.06 | 93.20 | **96.82**±0.1 |
| AA(%) | 91.66 | 90.07±3.9 | 94.64±0.7 | 95.03±0.2 | 95.38 | 95.19 | 93.13 | **96.64**±0.1 |
| Kappa(%) | 90.05 | 87.82±4.3 | 93.70±0.8 | 94.14±0.2 | 94.18 | 94.59 | 92.39 | **96.27**±0.1 |
| Test Time(s) | **96.54** | 128.45±15.3 | 230.62±11.7 | 352.30±33.1 | 454.21 | 168.22 | 346.86 | 412.18±28.8 |

is lower than our AHF-Net. Moreover, the feature fusion of ACO-SS+DBAF-Net uses the equal-weight strategy (element-wise adding and feature concatenating), so it cannot handle the multiscale terrain problem of remote sensing images well.

*5) Test Time:* Finally, we also compare the efficiency of all algorithms. Compared with ReuseNet, SML-CNN, and ACO-SS+DBAF-Net, our running time is the longest, but it is not much different from ACO-SS+DBAF-Net. Because AHF-Net uses unique fusion strategies CAFF and FPN-Net,

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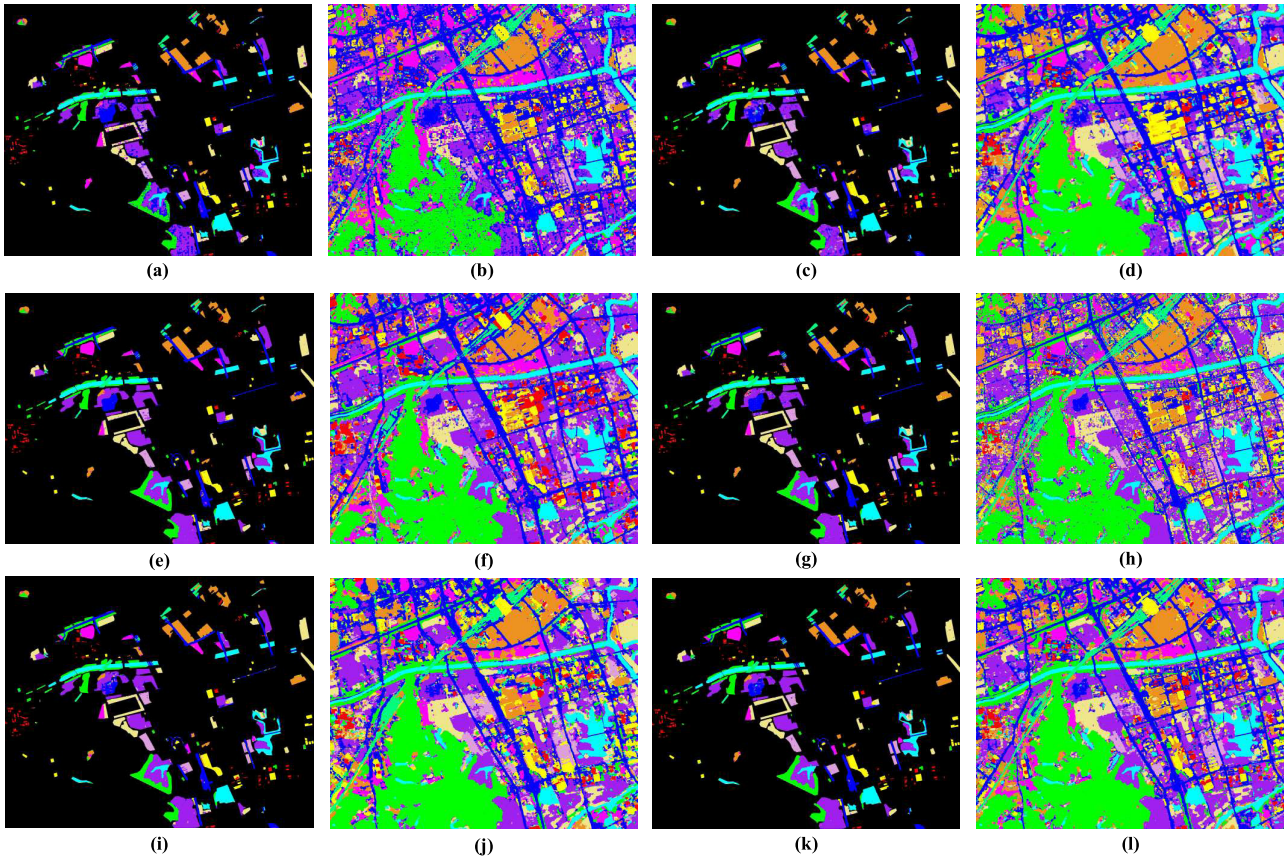



Fig. 8. Classification maps of different methods on the Nanjing Level 1A Images, each method has two classification maps: one is image with ground truth, while the other is overall image. (a) and (b) $\text{Net}_{\text{pansharp}}$. (c) and (d) FDBR18. (e) and (f) ACO-SS+DBAF-Net. (g) and (h) ReuseNet. (i) and (j) SML-CNN. (k) and (l) AHF-Net.

TABLE VIII
QUANTITATIVE CLASSIFICATION RESULTS OF NANJING LEVEL 1A IMAGES

| Methods | Net $_{pansharp}$ | FDBR18 | AWIHS+ FDBR18 | CAFF+ FDBR18 | ACO-SS+ DBAF-Net | ReuseNet | SML-CNN | AHF-Net |
|---|---|---|---|---|---|---|---|---|
| $c_1(Water)(\%)$ | 96.78 | 95.75±1.9 | 96.83±0.4 | 97.19±0.2 | 96.97 | 96.54 | 95.87 | **97.02**±0.2 |
| $c_2(Road1)(\%)$ | 80.94 | 81.75±3.8 | 91.53±1.3 | 91.97±0.9 | 93.69 | 92.67 | 89.35 | **93.51**±0.5 |
| $c_3(Bareland)(\%)$ | 91.68 | 94.19±3.1 | 96.77±1.1 | 96.73±0.9 | **98.36** | 96.24 | 95.97 | 97.23±0.4 |
| $c_4(Vegetation)(\%)$ | 94.59 | 92.88±4.1 | 94.61±0.8 | 95.59±0.6 | 97.16 | 90.76 | 95.84 | **97.55**±0.5 |
| $c_5(Building1)(\%)$ | 92.37 | 98.17±3.7 | 97.02±0.7 | 97.05±0.6 | 96.28 | 94.80 | 96.21 | **97.26**±0.5 |
| $c_6(Building2)(\%)$ | 99.45 | 94.62±1.7 | 98.05±1.4 | 98.75±1.5 | 98.12 | 97.77 | 99.41 | **99.45**±0.3 |
| $c_7(Building3)(\%)$ | 90.54 | 95.23±2.5 | 96.04±0.5 | 95.44±0.6 | **97.59** | 97.05 | 93.60 | 95.89±0.4 |
| $c_8(Building4)(\%)$ | 95.65 | 83.19±3.2 | 97.95±0.4 | 97.46±0.6 | **99.16** | 94.32 | 93.36 | 97.98±0.7 |
| $c_9(Building5)(\%)$ | 95.38 | 94.63±1.9 | 96.81±1.0 | 95.78±1.3 | 95.36 | 92.78 | 94.15 | **97.19**±0.8 |
| $c_{10}(Road2)(\%)$ | 96.51 | 84.17±3.5 | 95.72±1.1 | 95.72±1.6 | 97.02 | 92.76 | 97.92 | **97.50**±0.7 |
| $c_{11}(LowVegetation)(\%)$ | 77.55 | 91.12±2.7 | 93.84±1.1 | 92.72±1.6 | 75.23 | 91.43 | 86.36 | **94.48**±0.8 |
| OA(%) | 89.04 | 91.12±1.7 | 95.17±0.2 | 95.62±0.5 | 95.71 | 95.58 | 93.64 | **96.13**±0.1 |
| AA(%) | 91.95 | 91.78±1.8 | 95.37±1.3 | 95.86±0.7 | 94.99 | 95.71 | 93.24 | **96.46**±0.2 |
| Kappa(%) | 87.32 | 89.77±2.1 | 94.58±0.3 | 94.92±0.6 | 95.03 | 94.89 | 92.62 | **95.51**±0.1 |
| Test Time(s) | **102.3** | 138.36±9.4 | 157.59±9.9 | 338.46±11.4 | 306.15 | 151.12 | 226.03 | 423.41±35.7 |

the structure is more complicated. At the beginning of the network design, our starting point is to enable the network to obtain a powerful feature fusion capability while maintaining the feature extraction capability and improving classification accuracy. On the premise of keeping the main structure of AHF-Net unchanged, we also try to reduce the number of feature maps of each layer or the number of layers of the network, but the results are not satisfactory. Therefore, our next stage of research aims to maintain high accuracy while further improving the algorithm's efficiency.

*6) Comparison of Rest Three Data Set:* The comparison of the results of the other three data set is similar to the Xi'an Level 1A data set. For the three data sets, the AHF-Net obtains the highest AA, OA, and Kappa too.

For the Hohhot Level 1A data set, the quantitative and qualitative results are shown in Table VII and Fig. 7. It can

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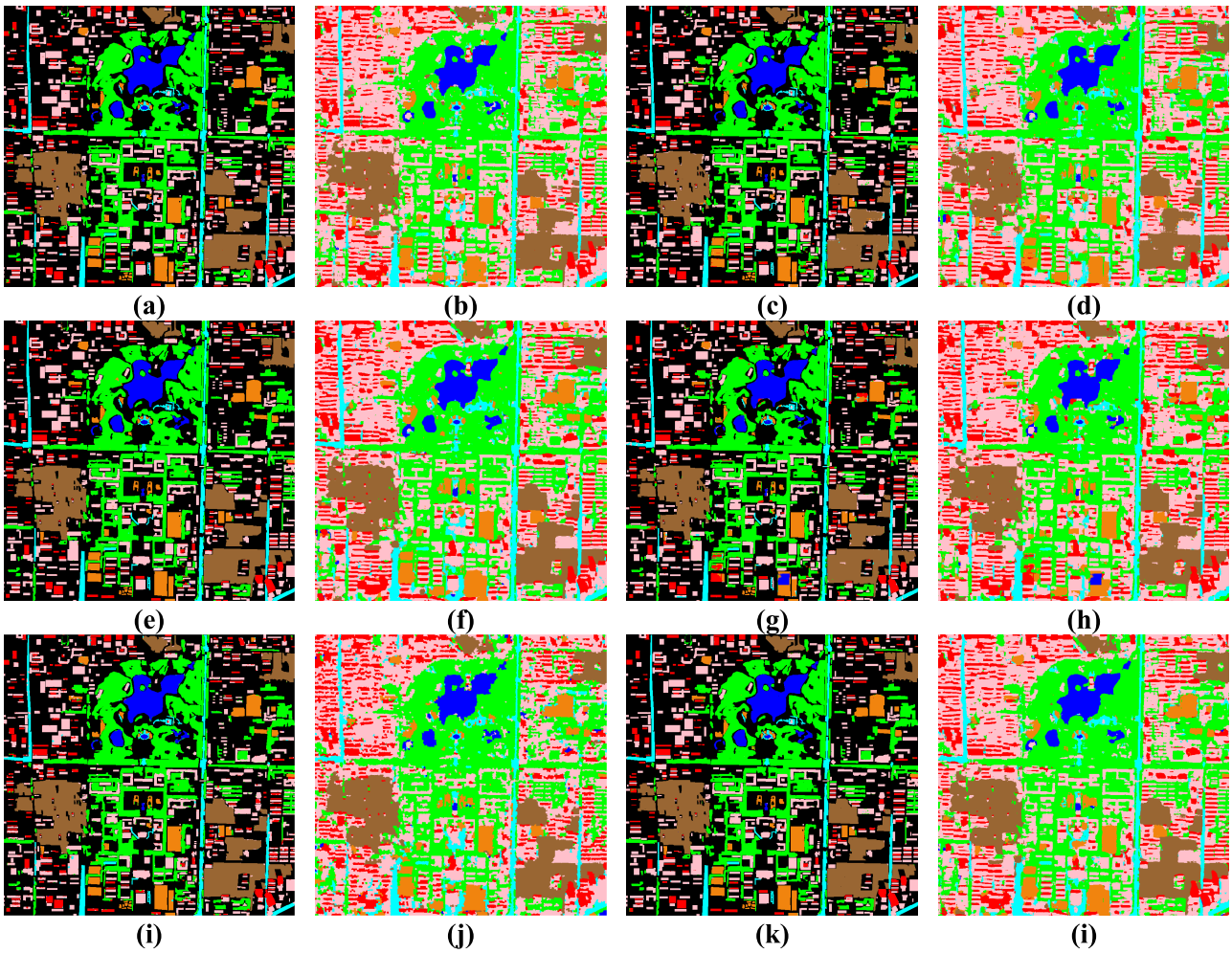



Fig. 9. Classification maps of different methods on the Xi'an Urban area Images, each method has two classification maps: one is image with ground truth, while the other is overall image. (a) and (b) $\text{Net}_{pansharp}$. (c) and (d) FDBR18. (e) and (f) ACO-SS+DBAF-Net. (g) and (h) ReuseNet. (i) and (j) SML-CNN. (k) and (l) AHF-Net.

be seen that c2 (Road1),c7 (Building3), and c9 (Building5) belong to more difficult categories in all categories. Through the analysis of ground truth and confusion matrix, we can see that c2 is widely distributed and intersects with multiple categories [c3 (Bareland), c5 (Building1), c6 (Building2)], so it is easily misclassified. This result indicates that we need to improve the discrimination ability of these class features through certain strategies. From the color of the overall images, we can see that the FDBR18 method results have some over-fitting. The accuracy of most categories of AHF-Net has reached the highest level, but some categories [c4 (Vegetation), c5(Building1), c11 (Building6)] have not achieved the best results. We consider that the possible reason is that data fusion leads to a reduction in the discriminative power of these types of features, thus reducing the classification results.

For the Nanjing Level 1A data set, the quantitative and qualitative results are shown in Table VIII and Fig. 8. We can see that c2 (Road), c4 (Vegetation), and c11 (Lowvegetation) belong to more difficult categories in all categories. Among them, c2 (Road1) and c10 (Road2), c4(Vegetation), and c11 (Lowvegetation) are similar in the spectrum and geographical shape. Moreover, these categories are widely distributed, and sample diversity is high. Our AHF net combines data fusion with feature fusion and uses a feature pyramid network to recognize multiscale terrain. Therefore, our AHF net achieved the highest OA, AA, and kappa. However, for the Nanjing level 1A data, our algorithm still has some categories that are not the most accurate [c3 (Bareland), c6 (building2), c7 (Building3)]. This may be because these categories' spectral characteristics are pronounced, but the increase in spatial details has caused AHF-Net to fall into the local optimum.

For the Xi'an Urban data set, the quantitative and qualitative results are shown in Table IX and Fig. 9. Because the data itself is small, ACO-SS+DBAF-Net has almost reached the upper-performance limit, so our AHF-Net improvement is not apparent. It can be seen that c2 (Flat-land) and c4 (Shadow) belong to more difficult categories. We can see that c2 is widely distributed and irregular in shape through the ground truth, while c4 (Shadow) and c1 (Building) are similar in the spectrum and easy to be misclassified. As mentioned above, our AWIHS can increase the information of MS and PAN images, reduce the difference between them, and

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TABLE IX
QUANTITATIVE CLASSIFICATION RESULTS OF XI'AN URBAN AREA IMAGES

| Methods | Net $_{pansharp}$ | FDBR18 | AWIHS+ FDBR18 | CAFF+ FDBR18 | ACO-SS+ DBAF-Net | ReuseNet | SML-CNN | AHF-Net |
|---|---|---|---|---|---|---|---|---|
| $c_1(Building)(\%)$ | 95.82 | 95.81±2.1 | 97.78±0.4 | 97.83±0.3 | 98.11 | 97.25 | 92.77 | **98.18**±0.2 |
| $c_2(Flat_land)(\%)$ | 92.44 | 92.00±6.4 | 96.99±0.7 | 98.32±0.4 | 99.91 | 98.12 | 95.29 | **99.92**±0.3 |
| $c_3(Road)(\%)$ | 89.69 | 96.44±2.7 | 96.93±0.9 | 96.40±0.6 | **99.90** | 96.04 | 92.25 | 99.21±0.8 |
| $c_4(Shadow)(\%)$ | 93.24 | 93.50±4.5 | 99.32±0.2 | 96.96±0.6 | 99.29 | 81.43 | 92.27 | **99.48**±0.06 |
| $c_5(Soil)(\%)$ | 93.82 | 98.44±1.8 | 98.37±0.2 | 99.32±0.1 | **99.81** | 99.28 | 98.81 | 99.57±0.4 |
| $c_6(Tree)(\%)$ | 94.05 | 97.62±0.9 | 99.02±0.2 | 98.13±0.2 | 98.43 | 98.46 | 97.66 | **99.13**±0.2 |
| $c_7(Water)(\%)$ | 96.45 | 97.50±1.8 | 98.12±0.1 | 98.81±0.4 | 99.61 | 93.32 | 97.35 | **99.66**±0.3 |
| OA(%) | 94.61 | 95.51±0.9 | 97.02±0.1 | 98.13±0.1 | 99.08 | 96.76 | 95.20 | **99.28**±0.1 |
| AA(%) | 94.48 | 95.90±1.2 | 97.13±0.2 | 97.97±0.1 | **99.29** | 94.84 | 95.58 | 99.20±0.2 |
| Kappa(%) | 92.38 | 95.50±1.2 | 96.17±0.1 | 97.59±0.2 | 98.88 | 95.81 | 94.69 | **99.09**±0.1 |
| Test Time(s) | **98.35** | 148.13±24.1 | 224.14±18.2 | 233.62±40.9 | 364.79 | 153.19 | 165.57 | 429.35±31.2 |

CAFF + FDBR18 Net can improve the discrimination of features and the recognition of different scale terrain. Our AHF-Net yields very high results, probably because each category's characteristics in the Xi'an Ubarn area are relatively simpler than those in the above data sets. Since the number of c3 (Road, 2.58%) data is relatively small, and the distribution of c5 (Soil) is too concentrated, the category c3 (Road), c5 (Soil) accuracy of our AHF-Net has not been improved to the highest (ACO-SS+DBAF-Net: c3: 99.90%, c5: 99.81%).

## V. CONCLUSION

In this article, we propose a novel AHF-Net for multiresolution remote sensing images classification. Experiments on several data sets have verified the effectiveness of our AWIHS strategy, CAFF module, and AHF-Net. However, our algorithm still has some shortcomings. First, it is not an end-to-end classification framework, and the deviations generated by AWIHS may affect subsequent classifications. Secondly, due to the relatively complex structure of AHF-Net, the computational complexity is relatively high, so it has a short-long running time. In the near future, on the premise of keeping the accuracy unchanged, we will focus on designing a more reasonable and effective classification network to improve multiresolution remote sensing image classification efficiency.

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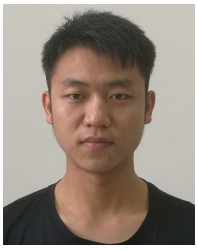
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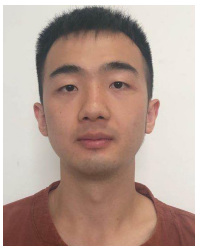
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